Bank Marketing Subscription Prediction

(Final QA)

Team 12

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Research Aims

1. Build a **predictive model** for term deposit subscription (Yes/No)
2. Explore **feature** importance for interpretability and marketing insights.
3. Compare algorithms
4. Ensure robust generalization (stratified k-fold CV)
5. Benchmark vs Kaggle baselines & related works



Problem Description

Research Problem:

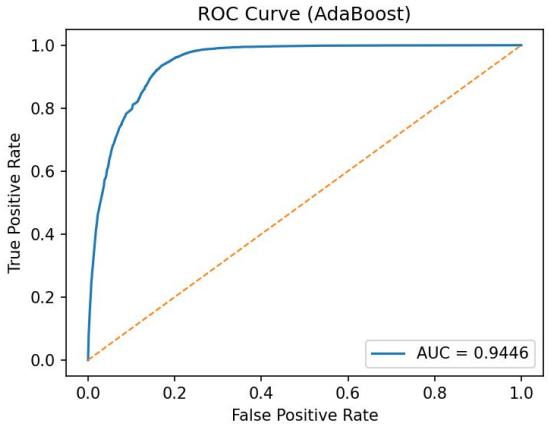
- This study aims to develop and evaluate interpretable machine learning models for predicting bank clients' subscription behavior.
- By addressing class imbalance and enhancing model explainability, the project seeks to improve targeting accuracy, reduce marketing costs, and provide actionable insights into key behavioral and demographic factors influencing client decisions.

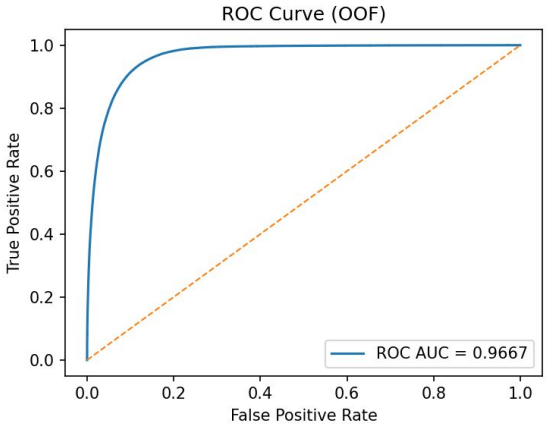
Dataset Overview

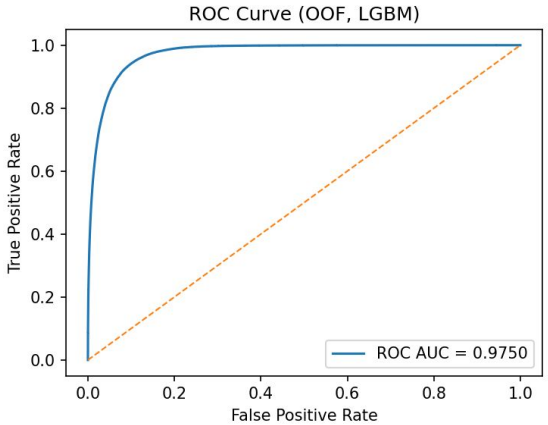
1. Dataset:
    - Kaggle Playground S5E8 (Bank Marketing Subscription)
    - https://www.kaggle.com/competitions/playground-series-s5e8/overview
2. Source:
    - Synthetic version of UCI Bank Marketing (Moro et al., 2014)
3. Size: 750k train, 250k test
4. Features:
    - Categorical: job, marital, education, housing, loan, contact type, etc.
    - Numerical: age, balance, duration, previous contacts, etc.
5. Target: subscribed (binary Yes/No)

ROC Curve Comparison

- AdaBoost < RNN < LGBM

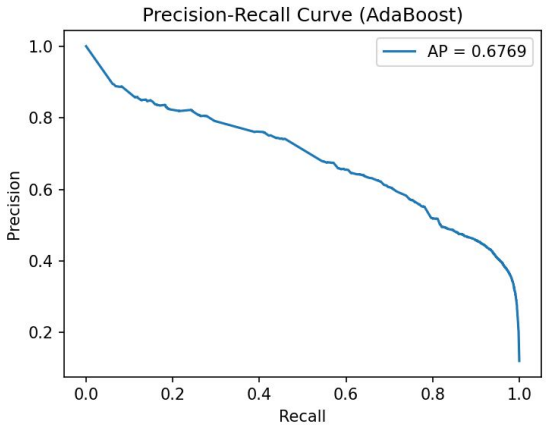


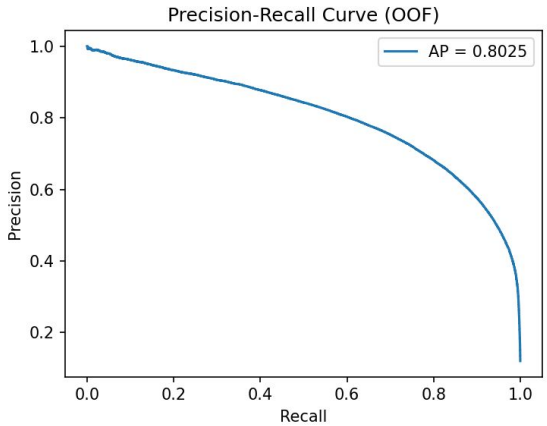


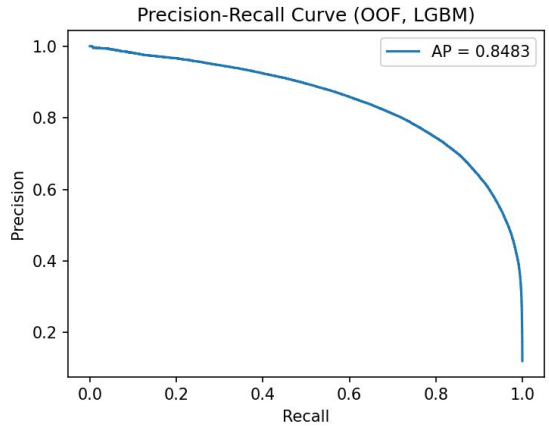


PR Curve Comparison

- AdaBoost < RNN < LGBM

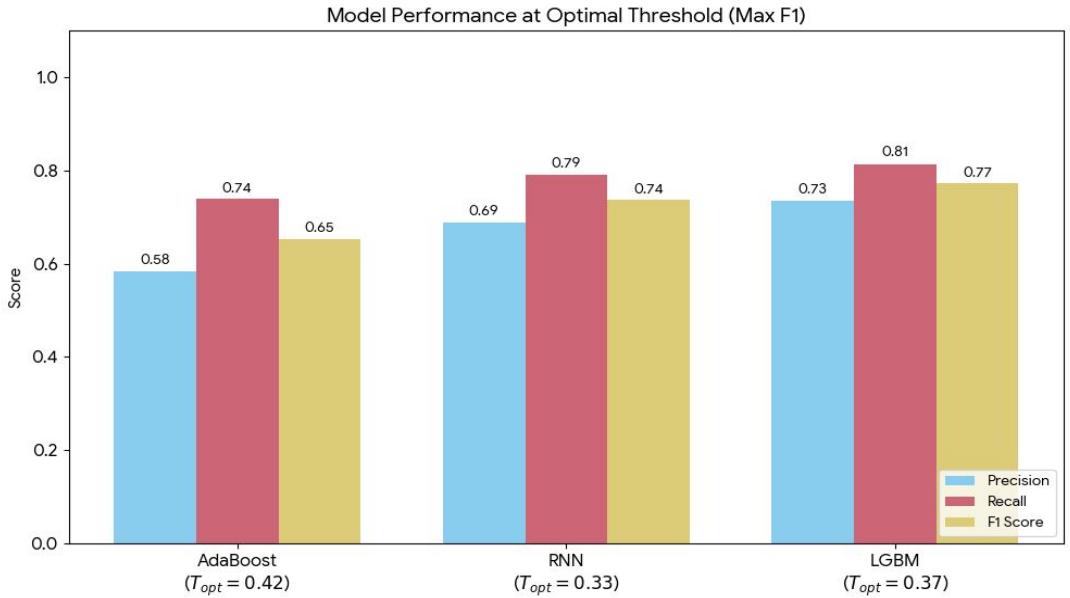






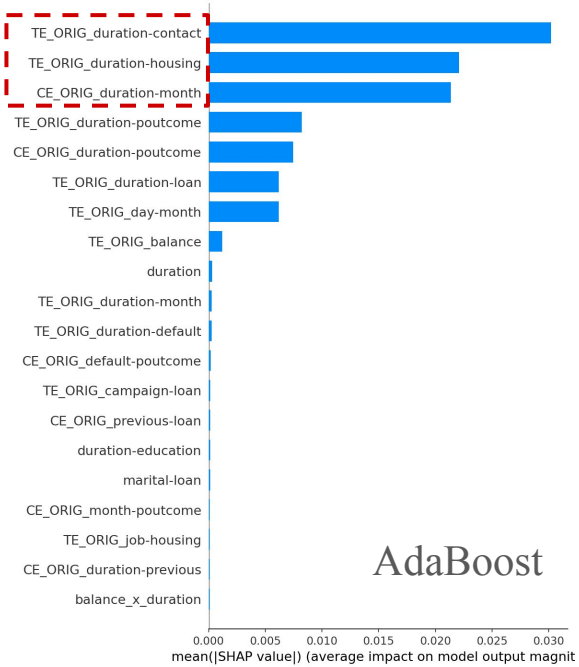
Optimal Threshold & F1 Analysis

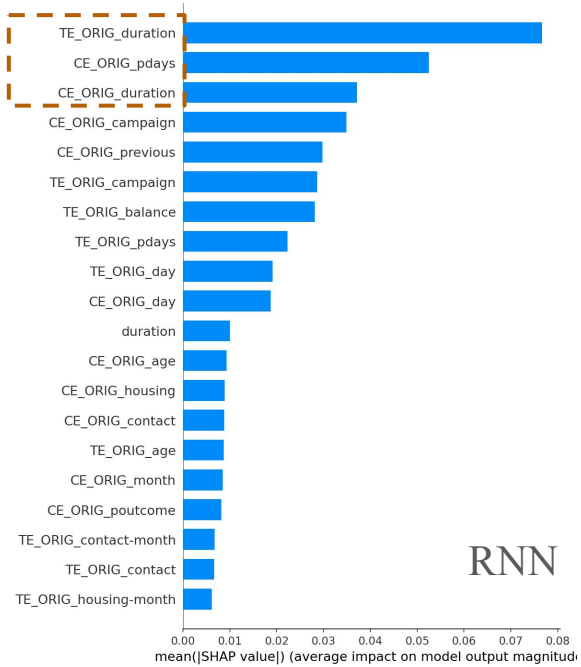
- AdaBoost < RNN < LGBM

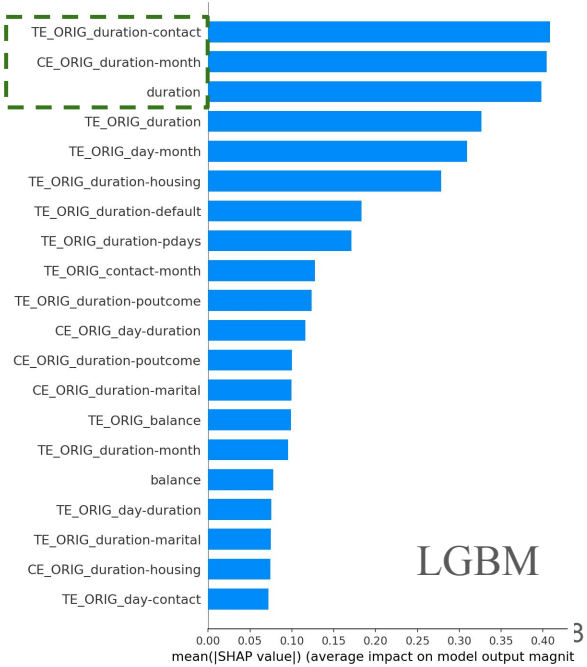


SHAP Feature Importance

- LGBM excels by balancing raw signals and interactions

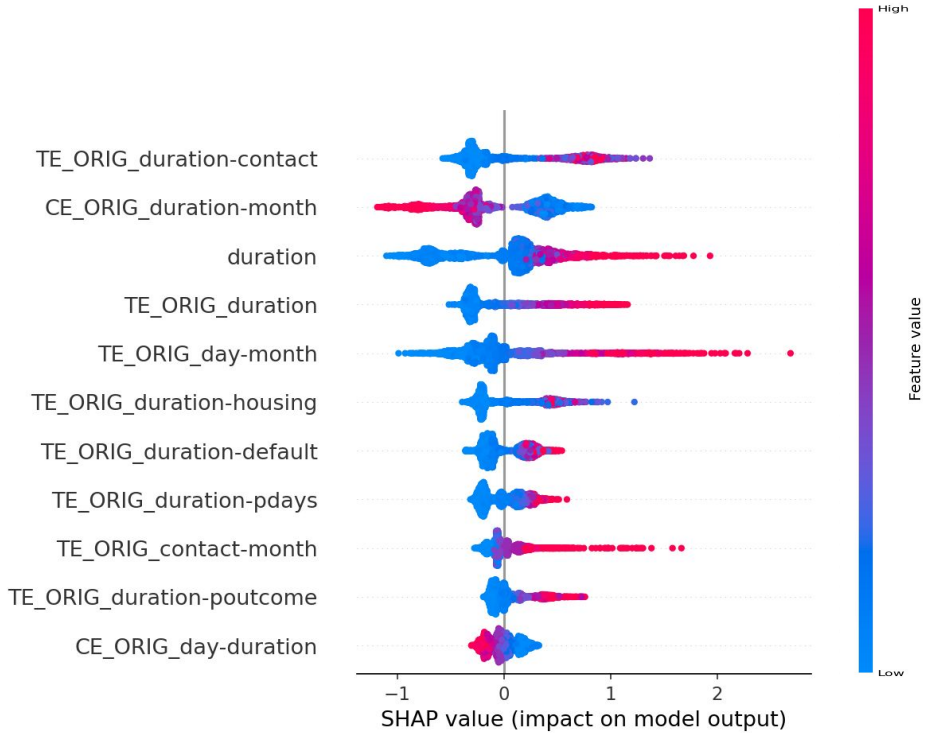






Decoding the Strategy

- Impact Direction
  - Higher values (Red), higher subscription probability
- Discrimination Power
  - Clear separation between potential subscribers and non-subscribers



Performance Summary Table

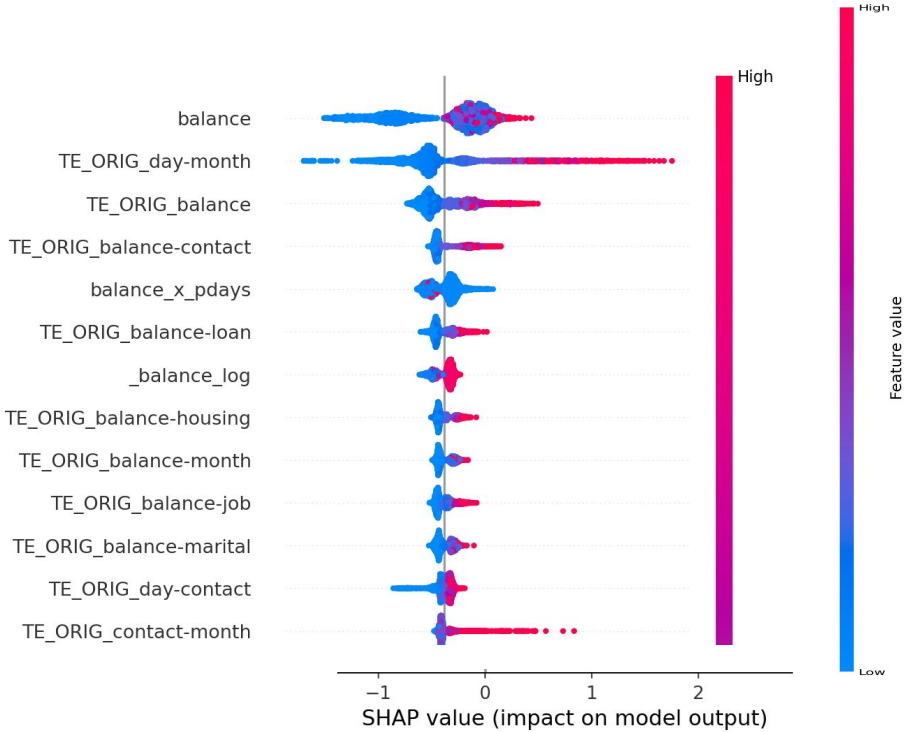
Model	F1-opt	Threshold	Precision	Recall	ROC-AUC	AP (PR-AUC)
AdaBoost	0.6517	0.421	0.5832	0.7384	0.9446	0.6769
RNN	0.7365	0.331	0.6888	0.7912	0.9667	0.8025
LightGBM	**0.7721**	0.368	**0.7345**	**0.8137**	**0.9750**	**0.8483**

Realistic Scenario: Predicting Without Call Duration

Metric	Original	**Without Duration**	Diff
F1 Score	0.7721	**0.5641**	↓ 27%
Threshold	0.368	**0.268**	↓ 0.1
ROC-AUC	0.9750	**0.7844**	↓ 0.19
AP (PR-AUC)	0.8483	**0.5404**	↓ 0.3

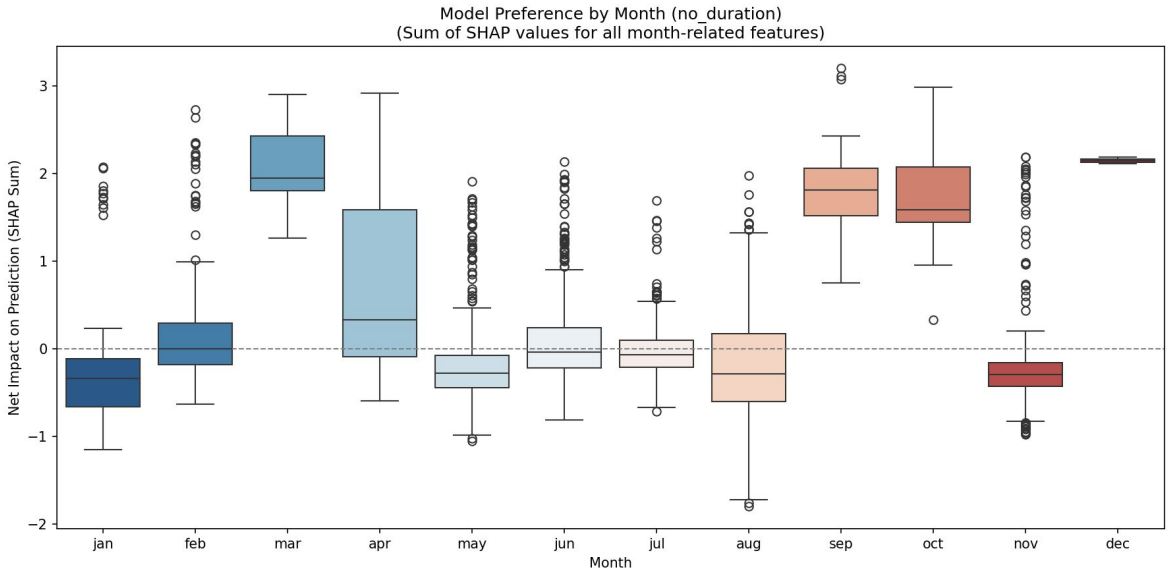
Strategic Pivot: What If We Don't Know Duration?

- Insight: Financial Power is the New King
  - Without call duration, Balance becomes the #1 predictor.
- Action: Prioritize "High Balance" Leads
  - Filter and rank calling lists by customer's financial depth before dialing.



Strategic Pivot: What If We Don't Know Duration?

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Insight

- Low Impact:
  - Static Demographics: Job, Marital, Education
  - Generic Loan Flags: Housing, Loan
  - Insight: Don't waste marketing resources on filtering lists simply based on "occupation"
- High Potential:
  - Financial Flow: Payday, large fund transfers
  - Digital Intent: Fixed deposit page browsing and interest rate calculation records.
  - Detailed feedback: A detailed code indicating the reason for rejection

Questions

Question 1

1. 請問有試圖去了解為何那特定三個月的訂 閱特別高的原因嗎？

Answer:

- 在 3, 9, 10, 12 月的訂閱率特別高, 推測跟消費者的財務管理習慣有關
- 財務週期性行為
  - 3 月 : 年終獎金發放後, 可支配資金增加 → 有餘裕做理財配置
- 消費淡旺季切換
  - 9–10 月 : 暑假旅遊旺季結束, 消費減少 → 進入儲蓄調整期
- 年底財務結算與銀行活動
  - 12 月 : 年度財務盤點、銀行年終優惠增加訂 閱誘因

Question 2

2. 若移除duration可能會以哪些策略或方向來去進行實驗以獲得好的效能

Answer:

模型必須依靠更多「弱訊號的結合」

- 使用更深的 boosting 模型（增加 num_leaves, max_depth）
- 使用 GBDT + 高等 embedding 模型（TabTransformer / TabM）
- 模型集成（RNN + LGBM + CatBoost）疊加弱訊號

Question 3

3. P.18你們提到說只在training set使用SMOTE跟5-fold cross, 我想了解說這兩個步驟為什麼能解決overfitting的問題跟提升generalization?

Answer:

- SMOTE（只用在 Training Set）
  - SMOTE 用來減少類別不平衡造成的「隱性過擬合」, 讓模型不會只記住少數正類樣本。
  - 在 training fold 中合成少數類樣本, 可使 decision boundary 更平滑、提升模型辨識少數類的能力。
  - 不在 validation / test 使用 SMOTE, 可避免資料洩漏, 確保模型評估是真實的。
- Stratified 5-Fold Cross-Validation
  - CV 的目的不是避免過擬合, 而是檢 查模型在不同資料切分下的穩定度。
  - 若 5 個 fold 的表現一致, 代表模型沒有依賴某一特定的訓練切分, 泛化能力較強。
  - 多次驗證可降低 variance, 提供比單次 Train–Test 更可靠的 generalization 估計。
  - Validation folds 保持原始不平衡比例, 因此能真實反映模型在部署時的表現。

Question 4

4. 報告提到使用了 Stratified k-fold CV 來確保泛化能力。但考慮到銀行定存數據通常具有強烈的『時間序列』特性（如報告中提到的 3, 9, 10, 12 月份效應），因此在預測未來的資料時表現可能會不穩定。想問你們有沒有考慮過 Out-of-time validation？或是你們對於如何處理這種未來資料時的會遇到的問題有什麼想法？

Answer:

- Kaggle S5E8 的月份屬於 類別型季節性特徵，並非連續的真實時間序列，因此無法直接採用Out-of-Time split
- 我們使用 Stratified k-fold CV 來檢查模型在不同資料切分下的穩定度，確保整體泛化能力。
- 若模型部署於真實銀行場景（具有真實時間流動），我們建議採用：
    - Time-based validation（Out-of-Time）
    - 定期再訓練（Re-training）
    - 處理 Concept Drift 的方法
- 本案中月份效應代表季節性，而非時間依賴，因此CV 是較合適的評估方式。

Question 5 6 7 8

5. 想問是如何具體設計這些交互特徵的？這些交互特徵是透過何種機制(例如, 簡單乘積、多項式擴展, 或是針對特定特徵的人工組合)來建立的？
6. 另外, 想麻煩你們具體列出所有的交互特徵, 以及簡單 說明為什麼設計這些交互特徵
7. 交互特徵是怎麼決定的, 有沒有有效找出來的方法？
8. 製作interaction features的具體做法是？ 如何強迫沒有相關聯的變項具有關聯性？

Answer:

- 1. 單變數非線性變換 (cos/sin + log)　　ex. _duration_sin, _duration_cos
- 2. 對所有數值欄位兩兩相乘　　ex. age_x_balance, duration_x_campaig
- 3. 對類別+數值的組合做「字串拼接 」　　ex. contact-month, duration-contact
- 4. 以上特徵再 Target / Count Encoding

Basic Exploratory Data Analysis (EDA)

Variable	Description	Type / Values
age	Client’s age	Numeric
balance	Average yearly balance (EUR)	Numeric
day	Last contact day of month	Numeric (1–31)
duration	Last contact duration (seconds)	Numeric
campaign	Number of contacts in this campaign	Numeric
pdays	Days since last contact (-1 = never contacted)	Numeric
previous	Number of contacts before this campaign	Numeric
y	Subscribed to term deposit? (target)	Binary: yes, no

Basic Exploratory Data Analysis (EDA)

Variable	Description	Type / Values
job	Type of job	Categorical: admin., blue-collar, entrepreneur, etc.
marital	Marital status	Categorical: married, single, divorced
education	Education level	Categorical: primary, secondary, tertiary, unknown
default	Has credit in default?	Categorical: yes, no
housing	Has a housing loan?	Categorical: yes, no
loan	Has a personal loan?	Categorical: yes, no
contact	Contact communication type	Categorical: unknown, telephone, cellular
month	Last contact month	Categorical: jan, feb, …, dec
poutcome	Outcome of previous campaign	Categorical: unknown, other, failure, success

Question 5 6 7 8

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8. 製作interaction features的具體做法是？ 如何強迫沒有相關聯的變項具有關聯性？

Answer:

- 我們並非強迫沒有關聯的變項 產生關聯, 而是先把可能的 candidate 顯式化
- 再由 Target Encoding 的統計結果與模型訓練去決定哪些組合真的有貢獻
- 並用正規化與交叉驗證控制過擬合

Question 9

9. target enocding和count encoding會用到什麼欄位上, 為什麼？

Answer:

- 大規模交互特徵後, 數 值會非常高
- 若直接做 one-hot：
  - 維度爆炸、記憶體爆炸
  - 模型非常容易 overfit
- TE / CE 剛好可以解決這些問題
  - 重新 encode 為合理分佈

Question 10

10. 為何使用target encoding可以解決類別不平衡的問題？實驗結果是否可證明 target encoding可以解決類別不平衡的影響？

Answer:

- 原本只知道某客戶是屬於哪類別
- 使用 TE 之後可以知道該類別平均訂 閱率為多少

- 例如平均訂 閱率為 20 %, 若可以知道 1% 的 student 訂閱率為 100 %
- 則會把 student 都預測為 1
- 而非把 1% 的 student 當作雜訊

Question 11

11. Preprocessing Steps, 寫到轉換類別變數至類別的意義不太清楚, 且為何能 夠解決模型泛化問題？此外, 是否從實驗效能可證明類別變數轉換成類別的策略, 可增加泛化性？

Answer:

- 避免模型把類別變成「假數 值」來處理 (job=1,2,3,4,5)
- 不轉成 category → 模型會基於「類別 ID 的大小」做切割

- 我們會補做實驗拿掉這部分試試看

Submission and Description	Private Score	Public Score	Selected
lgbm_no_duration_submit_f1thr.csv Complete (after deadline) · 5d ago · [LGBM-no_duration] CV AUC: 0.876079 Per-fold AUC: ['0.87960', '0.87499', '0.87...	0.87834	0.87583	☐
rnn_submit_f1thr.csv Complete (after deadline) · 8d ago · Per-fold AUC: ['0.96783', '0.96620', '0.96682', '0.96668', '0.96652'] [METRIC] O...	0.96837	0.96862	☐
ada_submit_f1thr.csv Complete (after deadline) · 8d ago · adaboost threshold,precision,recall,f1 0.42137032747268677,0.58324968358573...	0.94384	0.94422	☐
submit_f1thr.csv Error (after deadline) · 8d ago · [RNN] CV AUC: 0.966688 Per-fold AUC: ['0.96783', '0.96620', '0.96682', '0.96668', '...			
lgbm_submit_f1thr.csv Complete (after deadline) · 14d ago	0.97553	0.97576	☐